

(%i1) **a:matrix([1,2],[3,4]);**

a
$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

(%i2) **a_11:(-1)^(1+1)*determinant(minor(a,1,1));**

a_11 4

(%i6) **a_12:(-1)^(1+2)*determinant(minor(a,1,2));**

a_12 -3

(%i7) **a_21:(-1)^(2+1)*determinant(minor(a,2,1));**

a_21 -2

(%i8) **a_22:(-1)^(2+2)*determinant(minor(a,2,2));**

a_22 1

(%i9) **determinant(a);**

(%o9) -2

(%i10) **adjoint(a);**

(%o10)
$$\begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

(%i11) **invert(a);**

(%o11)
$$\begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\left(\frac{1}{2}\right) \end{bmatrix}$$

(%i12) **b:matrix([4,5],[6,7]);**

b
$$\begin{bmatrix} 4 & 5 \\ 6 & 7 \end{bmatrix}$$

(%i13) **b_11:(-1)^(1+1)*determinant(minor(b,1,1));**

b_11 7

(%i14) **b_12:(-1)^(1+2)*determinant(minor(b,1,2));**

b_12 -6

(%i15) **b_21:(-1)^(2+1)*determinant(minor(b,2,1));**

b_21 -5

(%i16) **b_22:(-1)^(2+2)*determinant(minor(b,2,2));**

b_22 4

(%i17) **determinant(b);**

(%o17) -2

(%i18) **adjoint(b);**

(%o18)
$$\begin{bmatrix} 7 & -5 \\ -6 & 4 \end{bmatrix}$$

(%i19) **invert(b);**

(%o19)
$$\begin{bmatrix} -\left(\frac{7}{2}\right) & \frac{5}{2} \\ 3 & -2 \end{bmatrix}$$

(%i20) **c:matrix([1,2,3],[7,8,9],[3,4,5]);**

$$c = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \\ 3 & 4 & 5 \end{bmatrix}$$

```
(%i21) c_11:(-1)^(1+1)*determinant(minor(c,1,1));
```

```
c_11 4
```

```
(%i22) c_12:(-1)^(1+2)*determinant(minor(c,1,2));
```

```
c_12 -8
```

```
(%i23) c_13:(-1)^(1+3)*determinant(minor(c,1,3));
```

```
c_13 4
```

```
(%i24) c_21:(-1)^(2+1)*determinant(minor(c,2,1));
```

```
c_21 2
```

```
(%i25) c_22:(-1)^(2+2)*determinant(minor(c,2,2));
```

```
c_22 -4
```

```
(%i26) c_23:(-1)^(2+3)*determinant(minor(c,2,3));
```

```
c_23 2
```

```
(%i27) c_31:(-1)^(3+1)*determinant(minor(c,3,1));
```

```
c_31 -6
```

```
(%i28) c_32:(-1)^(3+2)*determinant(minor(c,3,2));
```

```
c_32 12
```

```
(%i29) c_33:(-1)^(3+3)*determinant(minor(c,3,3));
```

```
c_33 -6
```

```
(%i30) determinant(c);
```

```
(%o30) 0
```

```
(%i31) adjoint(c);
```

```
(%o31) 
$$\begin{bmatrix} 4 & 2 & -6 \\ -8 & -4 & 12 \\ 4 & 2 & -6 \end{bmatrix}$$

```

```
(%i32) invert(c);
```

expt: undefined: 0 to a negative exponent.

-- an error. To debug this try: debugmode(true);

```
(%i33) d:matrix([4,7,8],[9,10,11],[12,13,14]);
```

$$d = \begin{bmatrix} 4 & 7 & 8 \\ 9 & 10 & 11 \\ 12 & 13 & 14 \end{bmatrix}$$

```
(%i34) d_11:(-1)^(1+1)*determinant(minor(d,1,1));
```

```
d_11 -3
```

```
(%i35) d_12:(-1)^(1+2)*determinant(minor(d,1,2));
```

```
d_12 6
```

```
(%i36) d_13:(-1)^(1+3)*determinant(minor(d,1,3));
```

```
d_13 -3
```

```
(%i37) d_21:(-1)^(2+1)*determinant(minor(d,2,1));
```

```
d_21 6
```

(%i38) **d_22**:(-1)^(2+2)·determinant(minor(d,2,2));

d_22 -40

(%i39) **d_23**:(-1)^(2+3)·determinant(minor(d,2,3));

d_23 32

(%i40) **d_31**:(-1)^(3+1)·determinant(minor(d,3,1));

d_31 -3

(%i41) **d_32**:(-1)^(3+2)·determinant(minor(d,3,2));

d_32 28

(%i42) **d_33**:(-1)^(3+3)·determinant(minor(d,3,3));

d_33 -23

(%i43) **determinant(d)**;

(%o43) 6

(%i44) **adjoint(d)**;

(%o44)
$$\begin{bmatrix} -3 & 6 & -3 \\ 6 & -40 & 28 \\ -3 & 32 & -23 \end{bmatrix}$$

(%i45) **invert(d)**;

(%o45)
$$\begin{bmatrix} -\left(\frac{1}{2}\right) & 1 & -\left(\frac{1}{2}\right) \\ 1 & -\left(\frac{20}{3}\right) & \frac{14}{3} \\ -\left(\frac{1}{2}\right) & \frac{16}{3} & -\left(\frac{23}{6}\right) \end{bmatrix}$$

(%i46) **e:matrix([1,2,3,4],[5,6,7,8],[9,10,11,12],[7,7,3,3]);**

e
$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 7 & 7 & 3 & 3 \end{bmatrix}$$

(%i47) **e_11**:(-1)^(1+1)·determinant(minor(e,1,1));

e_11 -16

(%i48) **e_12**:(-1)^(1+2)·determinant(minor(e,1,2));

e_12 16

(%i49) **e_13**:(-1)^(1+3)·determinant(minor(e,1,3));

e_13 16

(%i50) **e_14**:(-1)^(1+4)·determinant(minor(e,1,4));

e_14 -16

(%i51) **e_21**:(-1)^(2+1)·determinant(minor(e,2,1));

e_21 32

(%i52) **e_22**:(-1)^(2+2)·determinant(minor(e,2,2));

e_22 -32

(%i53) **e_23**:(-1)^(2+3)·determinant(minor(e,2,3));

```

e_23      - 32
(%i54) e_24:(-1)^(2+4)·determinant(minor(e,2,4));
e_24      32
(%i55) e_31:(-1)^(3+1)·determinant(minor(e,3,1));
e_31      - 16
(%i56) e_32:(-1)^(3+2)·determinant(minor(e,3,2));
e_32      16
(%i57) e_33:(-1)^(3+3)·determinant(minor(e,3,3));
e_33      16
(%i58) e_34:(-1)^(3+4)·determinant(minor(e,3,4));
e_34      - 16
(%i59) e_41:(-1)^(4+1)·determinant(minor(e,4,1));
e_41      0
(%i60) e_42:(-1)^(4+2)·determinant(minor(e,4,2));
e_42      0
(%i61) e_43:(-1)^(4+3)·determinant(minor(e,4,3));
e_43      0
(%i62) e_44:(-1)^(4+4)·determinant(minor(e,4,4));
e_44      0
(%i63) determinant(e);
(%o63) 0
(%i64) adjoint(e);
(%o64) 
$$\begin{bmatrix} -16 & 32 & -16 & 0 \\ 16 & -32 & 16 & 0 \\ 16 & -32 & 16 & 0 \\ -16 & 32 & -16 & 0 \end{bmatrix}$$

(%i65) invert(e);
expt: undefined: 0 to a negative exponent.
-- an error. To debug this try: debugmode(true);

```